Imagine you are part of a data science team working for a retail store that wants to improve its sales strategy. The store has collected data on customer purchases, including:

- Customer demographics (age, gender, income)
- Purchase history (items bought, frequency of purchases)
- Store traffic data (time spent in the store, peak hours)

## Task

Your task is to analyze this data to identify patterns and make recommendations to enhance customer satisfaction and sales performance.

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1. Linear Algebra

   Matrix Representation: How can customer data be represented as matrices? Discuss the potential for using matrix operations to analyse trends.

   Dimensionality Reduction: Consider using techniques like Principal Component Analysis (PCA). How can this help in understanding customer segments? What dimensions (features) would you focus on, and why?

2. Basic Probability

   Event Probability: If the store offers a discount on a specific product, what is the probability that a customer will purchase it? Discuss how you would calculate this probability using historical data.

   Bayesian Inference: How might you use Bayesian probability to update your beliefs about customer preferences based on new sales data?

3. Descriptive Statistics

   Central Tendency: What measures of central tendency (mean, median, mode) would you calculate for customer age and income? Why is it important to consider different measures?

   Variability: Discuss the significance of measuring variability (range, variance, standard deviation) in customer purchase amounts. How does understanding variability assist in inventory management?

   Visualisations: Which types of visualisations (histograms, box plots, scatter plots) would best represent the data you have? How can these visual tools help in identifying trends or outliers in customer behavior?